

Software-engineering capabilities across generations

	REPL / shell		IPython shell		Jupyter Notebook		IDE-native notebooks	
	capability		capability		capability		capability	
Interactive exploration	absent		absent		absent		absent	
Rich inline output	absent		partial / via add-ons	partial	native		native	
Literate narrative	absent		absent		native		native	
Language independence	absent		absent		native		native	
Reproducible environments	absent		partial / via add-ons	partial	partial / via add-ons	partial	native	
Interactive debugging	partial / via add-ons	partial	partial / via add-ons	partial	partial / via add-ons	partial	native	
Automated testing	partial / via add-ons	partial	partial / via add-ons	partial	partial / via add-ons	partial	native	
Refactoring / language server	absent		absent		partial / via add-ons	partial	native	
First-class version control	partial / via add-ons	partial	partial / via add-ons	partial	partial / via add-ons	partial	native	
Real-time collaboration	absent		absent		partial / via add-ons	partial	native	